Bivariate Ehrhart reciprocity for preorder polytopes

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Abstract

We prove Conjecture 7.2 of Athanasiadis and Chapoton: the double Ehrhart polynomial of every finite preorder satisfies $\mathbb{E}_\tau(-u,-v) = (-1)^n \mathbb{E}_\tau(u-1, v-1)$. The proof translates interior lattice points by the all-ones vector and applies ordinary Ehrhart reciprocity on each positive integral ray.

Task 1. Full solution

Let τ be a preorder on a nonempty finite set E, with $|E| = n$. An order ideal I satisfies $b \in I$, $a \leq_\tau b \Rightarrow a \in I$. This definition permits equivalent distinct elements of the preorder. For $I \subseteq E$, write $\Delta_I = \operatorname{conv}(0, e_i : i \in I) \subseteq \mathbb{R}^E$, and let $U(e)$ denote the principal order filter generated by e. The two-parameter polytope of [1, Section 3.2] is

$$\mathcal{Q}_\tau(r,s) = r \sum_{e \in E} \Delta_{U(e)} + s\Delta_E, \qquad r, s \in \mathbb{Z}_{\geq 0}.$$

Summing once for each element is the same as weighting the summand for an equivalence class by its cardinality, as in the source. By [1, Proposition 3.3], its inequality description is

$$x_e \geq 0 \quad (e \in E), \qquad \sum_{e \in I} x_e \leq r|I| + s \quad (\varnothing \neq I \text{ an order ideal}). \tag{1}$$

It is a lattice polytope, being an integral Minkowski sum of lattice simplices. The polynomial $\mathbb{E}_\tau(u,v)$ of total degree n is characterized by

$$\mathbb{E}_\tau(r,s) = \#(\mathcal{Q}_\tau(r,s) \cap \mathbb{Z}^E) \quad (r, s \in \mathbb{Z}_{\geq 0});$$

its existence is established in [1, Proposition 7.1].

Theorem 1. *For every finite preorder τ of size n, the following is a polynomial identity:*

$$\boxed{\mathbb{E}_\tau(-u,-v) = (-1)^n \mathbb{E}_\tau(u-1, v-1).} \tag{2}$$

Lemma 1. *For positive integers r, s, translation by $-\mathbf{1}$, where $\mathbf{1} = (1)_{e \in E}$, is a bijection*

$$\operatorname{int} \mathcal{Q}_\tau(r,s) \cap \mathbb{Z}^E \longrightarrow \mathcal{Q}_\tau(r-1, s-1) \cap \mathbb{Z}^E.$$

Proof. The polytope $\mathcal{Q}_\tau(r,s)$ is full dimensional: it contains $[0,r]^E$ by (1). Every normal in (1) is nonzero. Thus its interior consists exactly of the points satisfying all those inequalities strictly. This remains true for redundant inequalities: equality with a valid nonzero normal cannot occur at an interior point, and strict satisfaction of a finite system gives an open neighborhood contained in the polytope.

For an integral point x, the strict inequalities say precisely

$$x_e \geq 1, \qquad \sum_{e\in I} x_e \leq r|I| + s - 1 \quad (\varnothing \neq I \text{ an order ideal}).$$

Set $y = x - \mathbf{1}$. The same conditions become

$$y_e \geq 0, \qquad \sum_{e\in I} y_e \leq (r-1)|I| + (s-1).$$

By (1), these are exactly the defining inequalities for $\mathcal{Q}_\tau(r-1,s-1)$. Both translations preserve the integer lattice, which proves the bijection, including $r = 1$ or $s = 1$. □

Proof of the theorem. Fix positive integers r, s, and put $P = \mathcal{Q}_\tau(r,s)$. For every nonnegative integer k the Minkowski definition gives $kP = \mathcal{Q}_\tau(kr,ks)$. Hence the ordinary Ehrhart polynomial $L_P(k) = \#(kP \cap \mathbb{Z}^E)$ satisfies

$$L_P(k) = \mathbb{E}_\tau(kr,ks)$$

as an identity of polynomials in k, since the two polynomials agree at all nonnegative integers. In particular this equality is valid at $k = -1$. Ordinary Ehrhart–Macdonald reciprocity, applicable to the full dimensional lattice polytope P, now gives

$$\mathbb{E}_\tau(-r,-s) = L_P(-1) = (-1)^n \#(\operatorname{int} P \cap \mathbb{Z}^E) = (-1)^n \mathbb{E}_\tau(r-1,s-1),$$

where the last equality is Lemma 1.

The difference between the two sides of (2) is a polynomial in u, v vanishing at every positive integer pair. For each fixed positive integer v, it has infinitely many zeros as a polynomial in u, so all its coefficients in u vanish at that v. Each such coefficient is a polynomial in v vanishing at all positive integers and is therefore identically zero. This proves (2) for arbitrary u, v. □

If the empty preorder is allowed, its polynomial is 1, and the identity holds for $n = 0$ as well. Values with zero or negative parameters in the theorem are polynomial evaluations; the proof does not require giving negatively scaled Minkowski sums a counting interpretation.

Attribution, scope, and verification

The polytope descriptions and the existence of the double Ehrhart polynomial are established results of Athanasiadis and Chapoton. The contribution here is the translation argument and its application to their Conjecture 7.2. Their Remark 7.3 already explains how this identity implies their conjectured descent formula. Ordinary Ehrhart reciprocity is a classical theorem; no multivariate reciprocity theorem is needed.

The exact verifier independently enumerates Minkowski sums and inequality-defined lattice-point sets for all labeled preorders on at most four elements. It also checks the interior translation, the full polynomial identity using the source's draconian formula, and the published five-element example. These finite computations supplement the proof for arbitrary size.

All 26 pages of the current primary were read. Bounded public checks found no earlier full claim for this target and no identified journal final. The related paper [2] proves support-enumerator duality, a different statement; its relevant scope was checked. Review is by the authoring agent only, without independent peer review or proof-assistant certification.

Tasks 2–5. Proposed follow-up directions

Task 2. Give a direct word bijection for the descent formula implied by this result and [1, Remark 7.3], avoiding polynomial reciprocity in that combinatorial interpretation.

Task 3. Classify multivariate Minkowski families with a fixed integral translation carrying all interior lattice points to a family with the parameters decreased by prescribed integers.

Task 4. Refine the translation by coordinate monomial weights and derive a reciprocity formula for multivariate generating functions.

Task 5. Determine the restrictions on the mixed coefficients of $\mathbb{E}_\tau$ obtained from the parity of $\mathbb{E}_\tau(u-\frac{1}{2}, v-\frac{1}{2})$ under $(u,v)\mapsto(-u,-v)$.

References

[1] C. A. Athanasiadis and F. Chapoton, *Polytopes and posets associated to preorders*, arXiv:2605.26916v1, 2026.

[2] Z. Dai, Q. Hou, Z. Liu, W. Thawinrak, and H. Wang, *Counting Lattice Points in Minkowski Sums of Cross Polytopes*, arXiv:2608.16037v2, 2026.